

Q1: Using the characteristics of indicial notation, simplify all the expressions below.

a. $\varphi = \varepsilon_{ijk}\varepsilon_{lmn}C_{il}C_{jm}C_{kn} :$

$$\varphi = 6 \det(C)$$

b. $\nabla \times \nabla \phi :$

$$\nabla \times \nabla \phi = 0$$

c. $\nabla \cdot (\nabla \times \vec{E}) :$

$$\nabla \cdot (\nabla \times \vec{E}) = 0$$

Q2: Considering $\alpha = A_{ij}x_i x_j$, $\beta = B_{ij}x_i x_j$ with $A_{ij} = A_{ji}$ and $B_{ij} = -B_{ji}$, calculate the following.

a. $\frac{\partial \alpha}{\partial x_m} :$

$$\frac{\partial \alpha}{\partial x_m} = 2A_{mi}x_i$$

b. $\frac{\partial \beta}{\partial x_m} :$

$$\frac{\partial \beta}{\partial x_m} = 0$$

c. $\frac{\partial^2 \alpha}{\partial x_m \partial x_l} :$

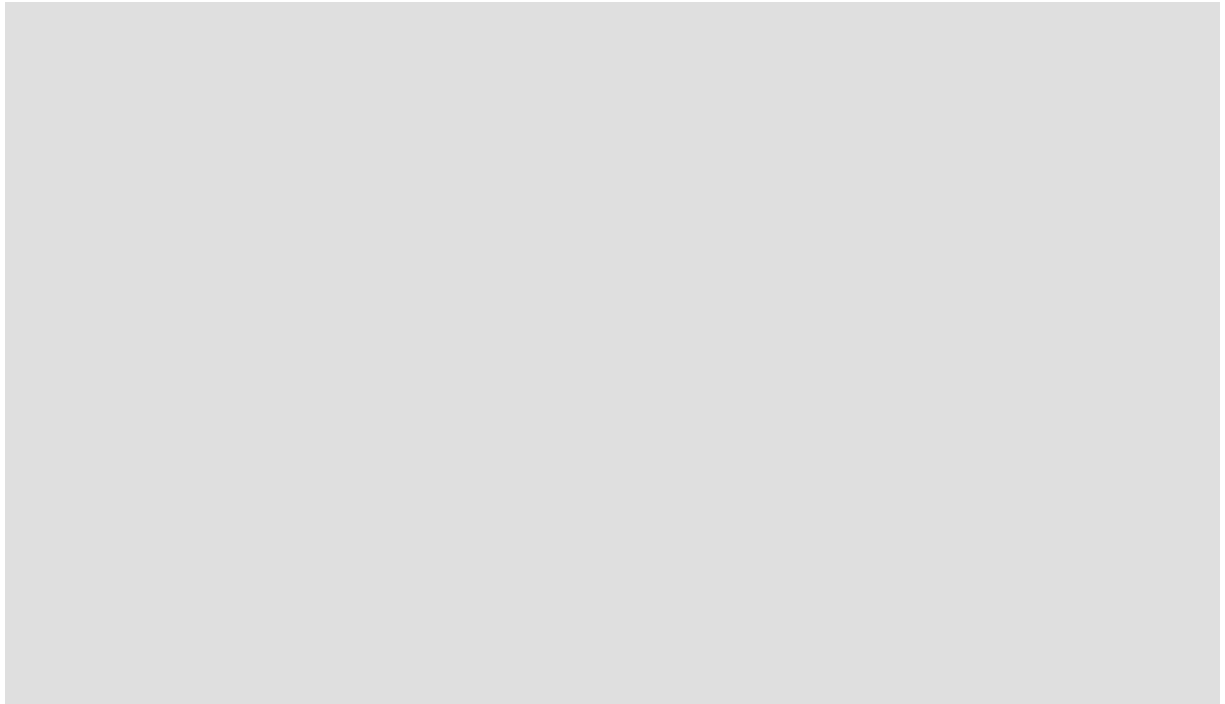
$$\frac{\partial^2 \alpha}{\partial x_m \partial x_l} = 2A_{lm}$$

d. $\frac{\partial^2 \beta}{\partial x_m \partial x_l} :$

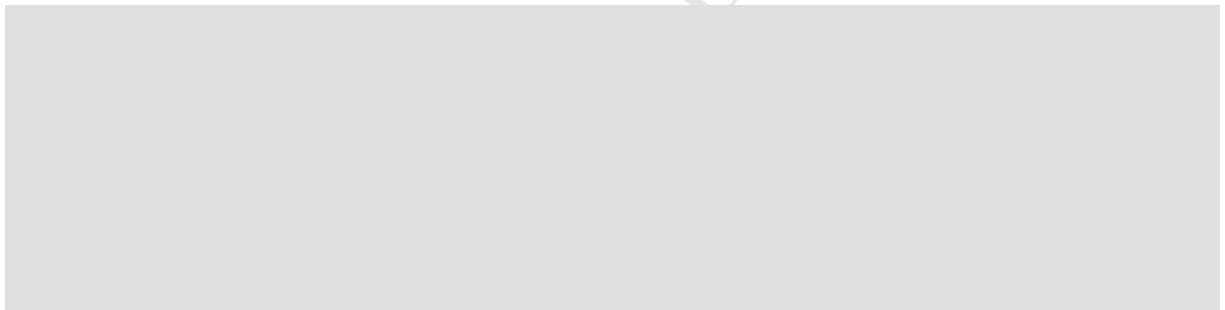
$$\frac{\partial^2 \beta}{\partial x_m \partial x_l} = 0$$

Q3: Prove the vector identities using indicial notation.

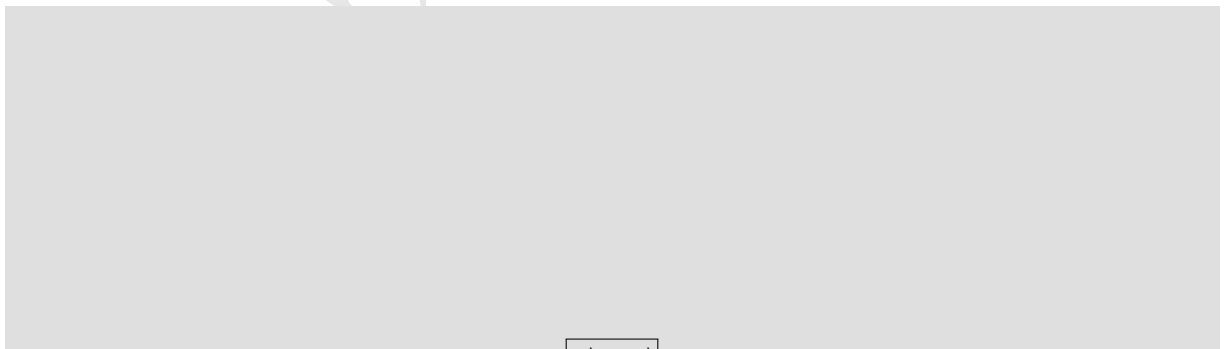
a. $\text{div}(\vec{u} \times \vec{v}) = (\text{curl } \vec{u}) \cdot \vec{v} - \vec{u} \cdot (\text{curl } \vec{v}) :$



b. $\nabla^2(\alpha\beta) = \alpha\nabla^2\beta + 2(\nabla\alpha) \cdot (\nabla\beta) + \beta\nabla^2\alpha :$



Q4: Let $\vec{v} = (\vec{a} \cdot \hat{n})\hat{n} + \hat{n} \times (\vec{a} \times \hat{n})$, where $\vec{a}$ is arbitrary and $\hat{n}$ is a unit vector. Express $\vec{v}$ in terms of the base vectors $\hat{e}_i$, expand and simplify.



$$\boxed{\vec{v} = \vec{a}}$$

Q5: Write $a_i = \frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j}$ in long form.

$$a_i = \frac{\partial v_1}{\partial t} + \frac{\partial v_2}{\partial t} + \frac{\partial v_3}{\partial t} + v_1 \left(\frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_1} + \frac{\partial v_3}{\partial x_1} \right) + v_2 \left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_2} \right) + v_3 \left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_3} \right)$$

Q6: Find the matrix of the tensor $\mathbf{T}$ which transforms any vector $\vec{a}$ into $\vec{b} = \vec{m}(\vec{a} \cdot \vec{n})$ where $\vec{m} = \frac{\sqrt{2}}{2}(\hat{e}_1 + \hat{e}_2)$ and $\vec{n} = \frac{\sqrt{2}}{2}(-\hat{e}_1 + \hat{e}_3)$.

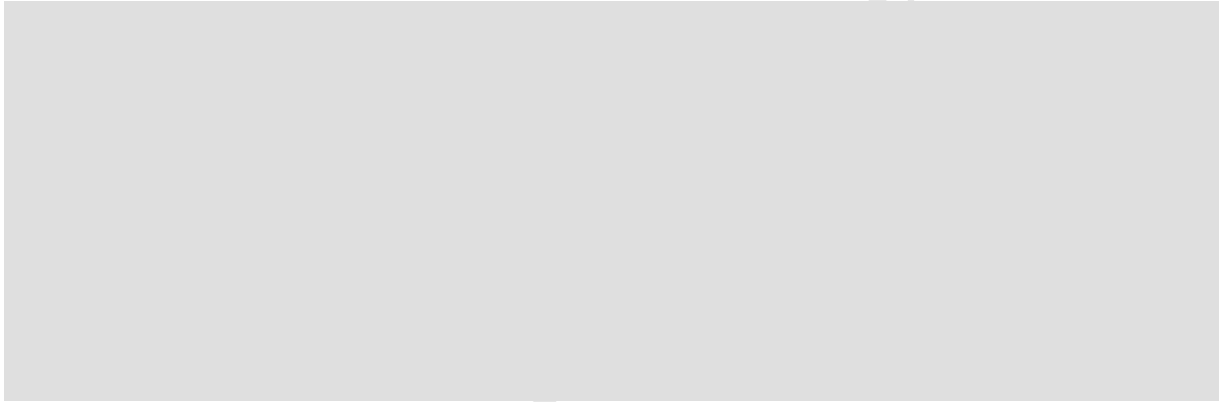
$$T = \begin{bmatrix} \frac{-1}{2} & 0 & \frac{1}{2} \\ \frac{-1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Q7: For $T_{ij} = \lambda E_{kk} \delta_{ij} + 2\mu E_{ij}$, $W = \frac{1}{2} T_{ij} E_{ij}$ and $P = T_{ij} T_{ij}$, show that

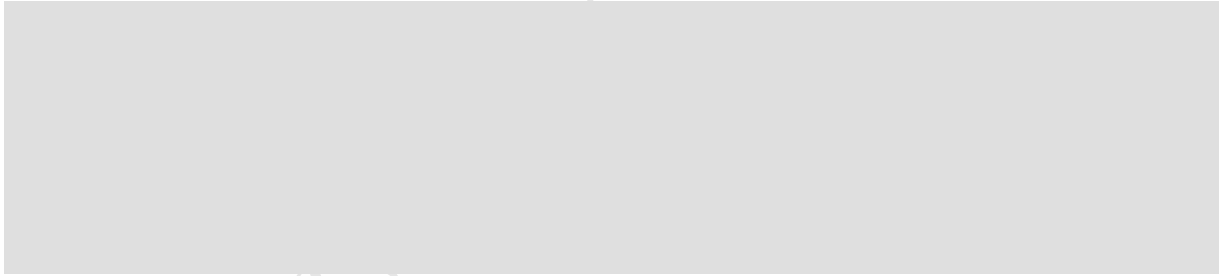
$$W = \mu E : E + \frac{\lambda}{2} [\text{Tr}(E)]^2$$

$$P = 4\mu^2 E : E + \lambda(3\lambda + 4\mu) [\text{Tr}(E)]^2$$

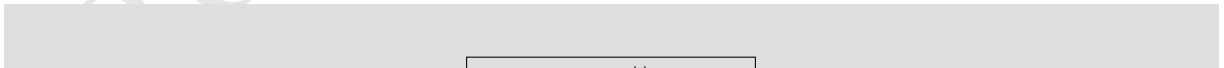
a. W :



b. P :



Q8: Find the invariant (vector form) of $\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}$.



$$\overleftrightarrow{\sigma} = \lambda \text{Tr}(\varepsilon) \overleftrightarrow{I} + 2\mu \overleftrightarrow{\varepsilon}$$

Q9: Using $\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}$, find strain-stress relationship's invariant form.

$$\overset{\leftrightarrow}{\varepsilon} = \frac{1}{2\mu} \overset{\leftrightarrow}{\sigma} - \frac{\lambda}{2\mu(3\lambda + 2\mu)} \text{tr}(\overset{\leftrightarrow}{\sigma}) \overset{\leftrightarrow}{I}$$

Q10: For the following vector field

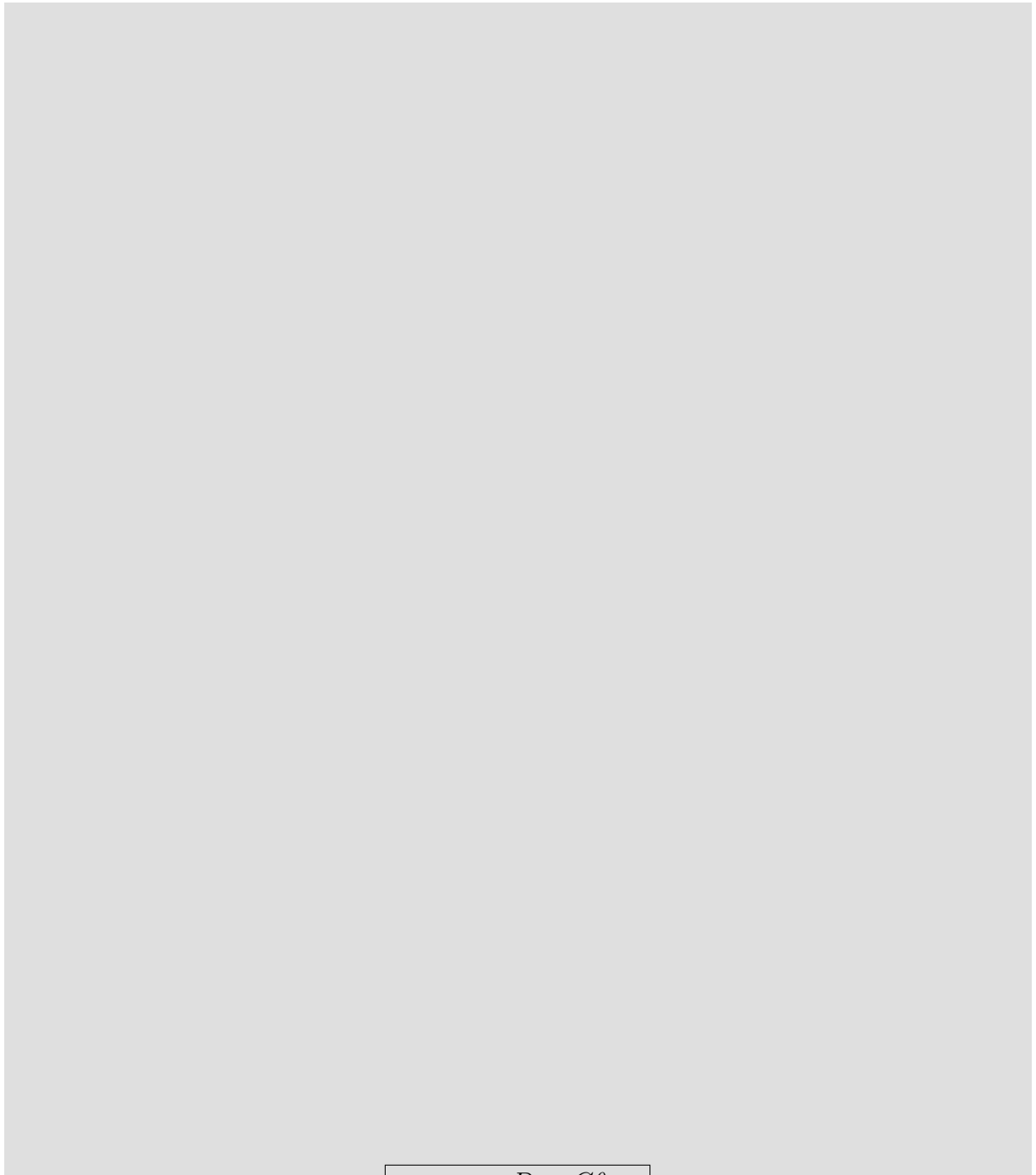
$$u_r = Ar + \frac{B}{r}, \quad u_\theta = C\theta + \frac{D}{r^2}, \quad u_z = z^2,$$

calculate a) $\nabla \vec{u}$, b) $\text{curl } \vec{u}$ and c) $\text{div } \vec{u}$ in cylindrical coordinates.

a. $\nabla \vec{u}$:

$$\nabla \vec{u} = \begin{bmatrix} A - \frac{B}{r^2} & -\frac{1}{r}(C\theta + \frac{D}{r^2}) & 0 \\ \frac{-2D}{r^3} & \frac{1}{r}(C + Ar + \frac{B}{r}) & 0 \\ 0 & 0 & 2z \end{bmatrix}$$

b. $\nabla \times \vec{u}$:



$$\nabla \times \vec{u} = \left(\frac{D}{r^3} - \frac{C\theta}{r} \right) \hat{e}_z$$

c. $\nabla \cdot \vec{u}$:

$$\nabla \cdot \vec{u} = 2A + \frac{C}{r} + 2z$$

Q11: Calculate a) the divergence and b) the curl of the tensor field

$$[A] = \begin{bmatrix} xy & y & z^2 \\ 1 & z & 0 \\ 0 & y & y \end{bmatrix}.$$

a. $\nabla \cdot \vec{A}$:

$$\nabla \cdot \vec{A} = \begin{bmatrix} 1 + y + 2z \\ 0 \\ 1 \end{bmatrix}$$

b. $\nabla \times \vec{A}$:

$$\nabla \times \vec{A} = \begin{bmatrix} 0 & 0 & x \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

Q12: Knowing that the displacement field is $\vec{u} = u_i \hat{e}_i$, and that the infinitesimal strain tensor is $\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i})$, find the components of the strain tensor in cylindrical coordinates.

$$\vec{\varepsilon} = \begin{bmatrix} \varepsilon_{rr} & \varepsilon_{r\theta} & \varepsilon_{rz} \\ \varepsilon_{\theta r} & \varepsilon_{\theta\theta} & \varepsilon_{\theta z} \\ \varepsilon_{zr} & \varepsilon_{z\theta} & \varepsilon_{zz} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_r}{\partial r} & \frac{1}{2}(\frac{\partial u_\theta}{\partial r} + \frac{1}{r}(\frac{\partial u_r}{\partial \theta} - u_\theta)) & \frac{1}{2}(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}) \\ \frac{1}{2}(\frac{\partial u_\theta}{\partial r} + \frac{1}{r}(\frac{\partial u_r}{\partial \theta} - u_\theta)) & \frac{1}{r}(u_r + \frac{\partial u_\theta}{\partial \theta}) & \frac{1}{2}(\frac{1}{r}\frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z}) \\ \frac{1}{2}(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}) & \frac{1}{2}(\frac{1}{r}\frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z}) & \frac{1}{2}(\frac{\partial u_r}{\partial r}) \end{bmatrix}$$